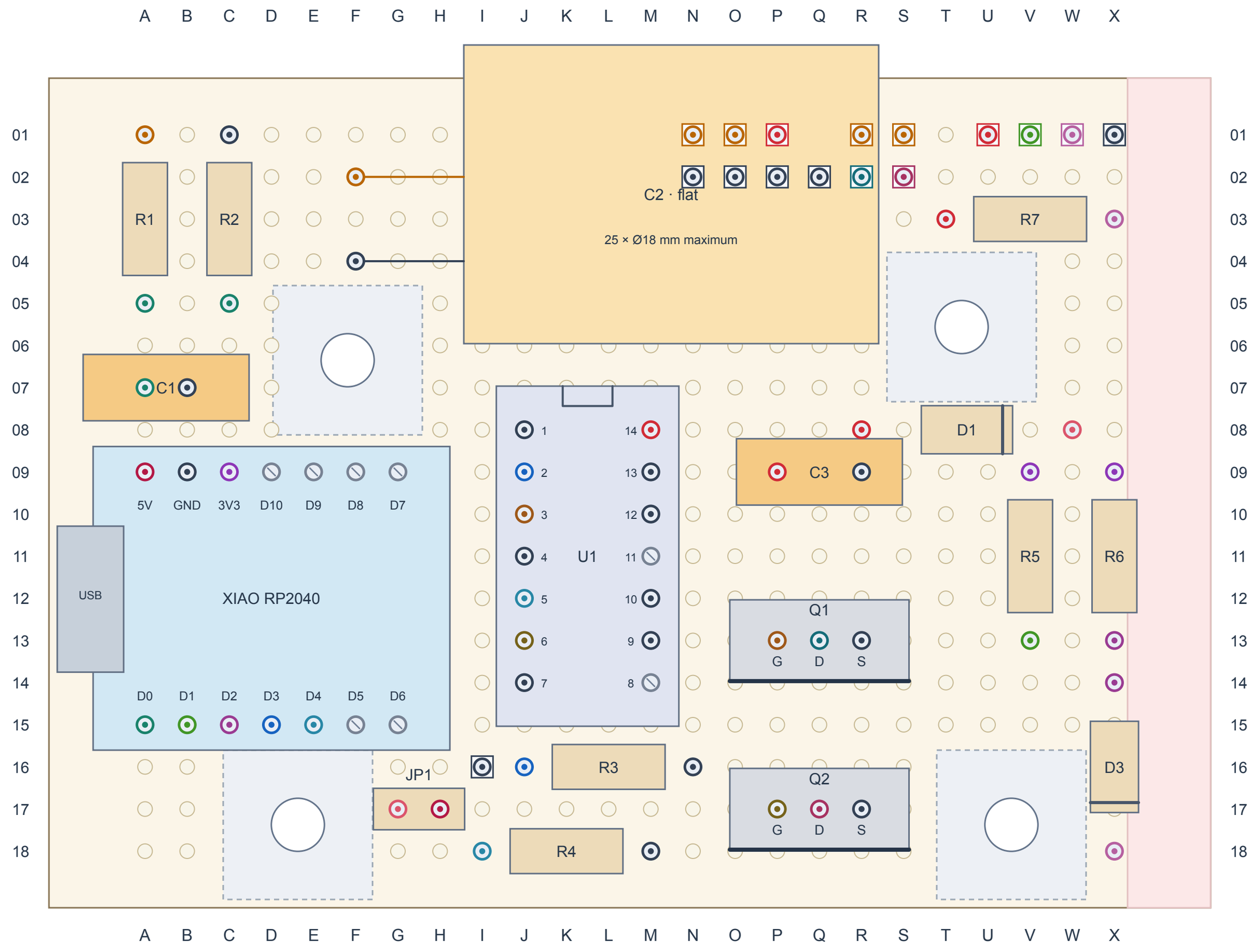
COMPONENT SIDE — USB at left

70 × 50 mm • 24 × 18 holes • 2.54 mm pitch • all point-to-point wires are on rear.

Solid dots: through-hole leads. Square ports: rear-only cable attachment pads.

Coordinates are assigned by this drawing; supplier lettering may be on the opposite face.

R1 100k · R2 22k · R3/R4 10k · R5/R6 4.7k · R7 3.3k
C1/C3 0.1 µF · D1/D3 NTE585 · JP1 removable shunt (remove for USB)
D2 stays at the pump; C4 stays at the Hall sensor. Neither mounts on this board.



U1 notch ↑ • Q1/Q2 printed faces ↓ • D1 stripe → W08 • D3 stripe → X18 • C2 + F02 / − F04

Grey squares: 9 × 9 mm mounting keep-outs. Centres (x,y): (18,17), (55,15), (15,45), (58,45) mm.

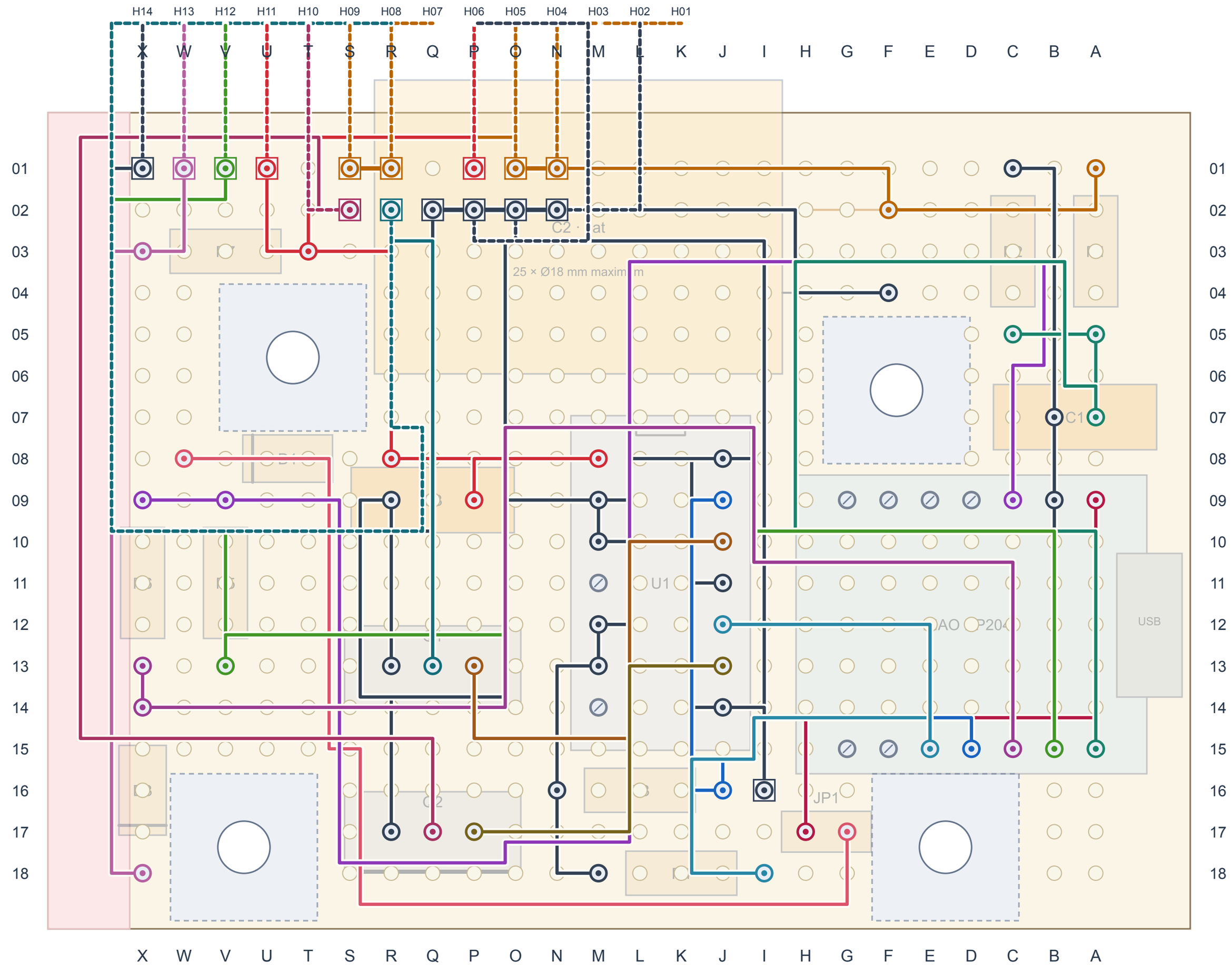
Grid offsets are nominal. Verify the board and dry-fit C2 before drilling or soldering. This is not a 1:1 drill template.

SOLDER SIDE — mirrored

Flip the board left ↔ right; top stays top. Pad names do not change.

Insulated wires cross without joining. Solder only at marked endpoints; bends are not joints.

Only W01, W02 and W05–W07 are short bare bus links. All other links are insulated.



Solder-side drawing shows every wire; use the wire schedule or interactive net filter to trace crowded areas.

Grey squares: 9 × 9 mm mounting keep-outs. Centres (x,y): (18,17), (55,15), (15,45), (58,45) mm.

Grid offsets are nominal. Verify the board and dry-fit C2 before drilling or soldering. This is not a 1:1 drill template.

CABLE CONNECTIONS AND COLOUR KEY

Dashed leads attach to rear copper pads. Last column: wire centre behind the board.

H01	N01	S1 switched output	2.7 mm
H02	N02	Battery negative / J1 sleeve	6.5 mm
H03	O01	Buck IN+	6.5 mm
H04	O02	Buck IN−	4.6 mm
H05	P01	Buck OUT+ (set to 5.0 V)	4.6 mm
H06	P02	Buck OUT−	8.4 mm
H07	R01	Pump + / D2 striped end	4.6 mm
H08	R02	Pump − / D2 unstriped end	8.4 mm
H09	S01	LED + (JST SM)	2.7 mm
H10	S02	LED − (JST SM)	2.7 mm
H11	U01	Lock +5 V: A3144 pin 1 / C4	2.7 mm
H12	V01	Lock HALL: A3144 pin 3	2.7 mm
H13	W01	Lock SW: microswitch NO	2.7 mm
H14	X01	Lock GND: Hall pin 2 / COM / C4	2.7 mm

OFF-BOARD CONNECTIONS

Battery + → J1 centre and S1 input. S1 output → H01. Battery − → J1 sleeve and H02.
Leave J1 switched contact unused. Set the buck output to 5.0 V before connecting H05/H06.
D2 NTE586 across pump terminals: stripe to pump +. C4 0.1 µF across A3144 pins 1–2.
A3144: printed face toward you, leads down; pins 1/2/3 = +5V/GND/HALL.
Twist switch COM → GND, NO → SW; NC unused. Lock cable uses the JST PH pair.
JP1: fit shunt for battery operation; remove for USB and follow build-guide power procedure.

- Switched battery
- Ground
- Regulated 5 V
- D1 → JP1
- JP1 → XIAO 5V
- 3.3 V
- Battery sense
- Hall signal
- Twist GPIO
- Switch contact
- Pump PWM
- LED PWM
- Pump gate
- LED gate
- Pump return
- LED return

Cable H02 attaches to N02. Capacitor C2 positive uses F02.

WIRE-BY-WIRE SOLDER SCHEDULE

Use component-side pad names on both faces. All links on rear; solder only at their two endpoints.

W01	N01 → O01	VLOAD	BARE link	W27	R08 → P09	5V_REG	4.6 mm deep
W02	R01 → S01	VLOAD	BARE link	W28	T03 → U01	5V_REG	2.7 mm deep
W03	O01 → R01	VLOAD	2.7 mm deep	W29	M08 → P09	5V_REG	2.7 mm deep
W04	F02 → N01	VLOAD	2.7 mm deep	W30	T03 → P01	5V_REG	6.5 mm deep
W05	N02 → O02	GND	BARE link	W31	T03 → R08	5V_REG	2.7 mm deep
W06	O02 → P02	GND	BARE link	W32	W08 → G17	AFTER_D1	2.7 mm deep
W07	P02 → Q02	GND	BARE link	W33	A09 → H17	XIAO_5V	2.7 mm deep
W08	F04 → N02	GND	2.7 mm deep	W34	V09 → X09	3V3	2.7 mm deep
W09	R13 → Q02	GND	2.7 mm deep	W35	C09 → V09	3V3	6.5 mm deep
W10	R17 → P02	GND	6.5 mm deep	W36	A05 → A07	ADC	2.7 mm deep
W11	I16 → O02	GND	2.7 mm deep	W37	A05 → C05	ADC	4.6 mm deep
W12	A01 → F02	VLOAD	4.6 mm deep	W38	A15 → A07	ADC	8.4 mm deep
W13	M10 → M09	GND	2.7 mm deep	W39	V13 → V01	HALL	2.7 mm deep
W14	M13 → M12	GND	2.7 mm deep	W40	B15 → V13	HALL	4.6 mm deep
W15	B09 → B07	GND	2.7 mm deep	W41	X13 → X14	TWIST	2.7 mm deep
W16	M12 → M10	GND	2.7 mm deep	W42	C15 → X14	TWIST	8.4 mm deep
W17	J14 → I16	GND	4.6 mm deep	W43	X03 → W01	SW	4.6 mm deep
W18	N16 → M18	GND	4.6 mm deep	W44	X03 → X18	SW	6.5 mm deep
W19	J08 → J11	GND	6.5 mm deep	W45	D15 → J16	PWM_P	8.4 mm deep
W20	J11 → J14	GND	4.6 mm deep	W46	J09 → J16	PWM_P	8.4 mm deep
W21	J08 → M09	GND	2.7 mm deep	W47	E15 → I18	PWM_L	6.5 mm deep
W22	M13 → N16	GND	2.7 mm deep	W48	E15 → J12	PWM_L	4.6 mm deep
W23	M09 → R09	GND	2.7 mm deep	W49	J10 → P13	GATE_P	4.6 mm deep
W24	C01 → B07	GND	2.7 mm deep	W50	J13 → P17	GATE_L	2.7 mm deep
W25	B09 → J08	GND	6.5 mm deep	W51	Q13 → R02	DRAIN_P	6.5 mm deep
W26	R09 → X01	GND	4.6 mm deep	W52	Q17 → S02	DRAIN_L	4.6 mm deep

Q02 and I16 are wire junction pads; no component goes in these holes.

Crossings and bends are not joints. Keep all other bare metal inside a solder fillet.